

▣ 27 Simple Regression

Recall example 2 from the chapter on correlation.

Example 2

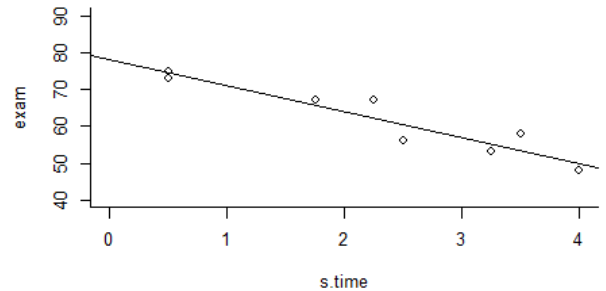
A study is carried out on a sample of ten children to investigate the relationship, if any, between screen time (the average time they spent per day using a smart phone / tablet) and their score in a diagnostic test.

s.time (hours)	0.5	1.75	2.5	2.25	3.5	3.25	4	0.5	0.5
exam (x 100)	75	67	56	67	58	53	48	73	75

The scatter plot suggests an inverse relationship.

We found that there was a statistically significant (negative) correlation (i.e. a straight-line relationship) between the two variables with $r = -0.95$

An obvious next step is to try and quantify that relationship (i.e. fit the straight line shown in the plot)



Recall from coordinate geometry, the equation of a line is given by $y = mx + c$

The convention in regression is to use **a** for intercept, which is sometimes called the constant and **b** for slope, which is sometimes called the coefficient (of s.time).

c (or a)? We can guess the intercept is about 80
m (or b)? The slope is the rise divided by the step.

The exam values range from about 80 to 50, giving a rise of about -30.
The s.time values range from about 0 to 4, giving a step of about 4
Thus, the slope is about $-30/4 = -7.5$

So, the equation of the line is about $y = -7.5x + 80$ or exams = $-7.5 \text{ s.time} + 80$

We can find the line parameters precisely using R. The coefficient (of s.time) is -7.1 and the constant is 78.3.

```
> lm(exam~s.time)
```

```
Call:
lm(formula = exam ~ s.time)
```

```
Coefficients:
(Intercept)      s.time
    78.250       -7.053
```

```
> abline(lm(exam~s.time)) # fits the regression line on the scatter plot
```

Such a construct is called a regression model.

The independent variable, s.time, is sometimes called a predictor or explanatory variable, and the dependent variable is sometimes called the response variable.

Interpreting the parameters

The coefficient?

The coefficient of s.time (-7.1) indicates the average effect on exam score per unit change in s.time.

i.e. every extra hour spent looking at a screen by a child will see a decline in their exam score of 7.1 marks.

E.g. children who spend say 4 hours looking at screens might expect to score 14.2 marks less (2×-7.1) than children who spend only 2 hours looking at screens.

The constant?

The constant (78.3) is the expected exam score for children who spend zero hours looking at screens.

In many cases, the constant does not have a meaningful interpretation.

(E.g. if we build a regression model to attempt to predict a person's weight given their height, it would not be sensible to use such a model to predict the weight when the height is zero!)

Prediction

One purpose of regression is to make predictions.

Example

What might be expected as an average exam score for children who spend 3 hours per day looking at screens?

Answer: exam = $-7.1 \text{ s.time} + 78.3 = -7.1 (3) + 78.3 = 57$

What about children who spend 10 hours per day looking at screens?

Answer: exam = $-7.1 \text{ s.time} + 78.3 = -7.1 (10) + 78.3 = 7.3$

Note that the last example required us to specify a value for s.time of 10, which exceeds any actual value for this variable observed (the largest s.time value was 4. Thus, we extrapolated

Caution needs to be exercised when extrapolated predictions. Interpolated predictions will be more trustworthy.

Example 1 (again)

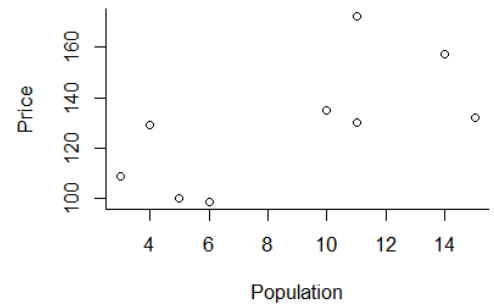
An analysis is carried out to determine how, if at all, the price of houses is related to population density in the towns in which they are located. A selling price for a number of houses that have recently sold are noted as are the population of the towns in which they are located and are tabulated below.

Population (000)	3	14	15	6	10	11	4	11	5
Price (000)	109	157	132	99	135	130	129	172	100

Over the range of the data, prices vary from about 100 to 160, a change of 60, and population varies from about 3 to about 15, a change of 12 so we might estimate the slope to be $60/12 = 5$

And the intercept is about 100 thus our regression equation might be

$$\text{price} = 100 + 5 \text{ population}$$



In fact, R shows that the equation is $\text{price} = 96.6 + 3.7 \text{ population}$

```
lm(Price~Population)
```

Call:

```
lm(formula = Price ~ Population)
```

Coefficients:

```
(Intercept)    Population
    96.581         3.719
```

Again, we could use this equation to make predictions.

Example

Predict the price of a house in a town of population 9,000.

Answer

$$\text{Population} = 96.6 + 3.7 (9) = 129.9 \text{ or } \text{€}129,900$$

Example 3 (again)

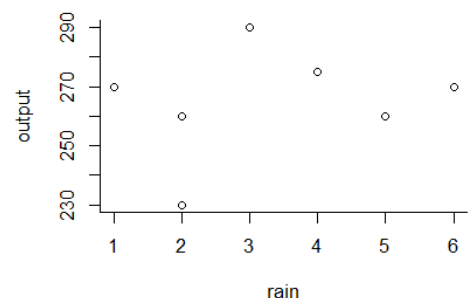
A study is carried out in a production facility to determine if inclement weather affects productivity. A random sample of shifts is taken from the last 6 months and the number of units manufactured in that shift observed. A meteorological database is consulted to determine the amount of rain that fell during that shift.

Rain	2	3	1	4	5	6	2
Output	260	290	270	275	260	270	230

Recall in this example we rejected the hypothesis that the correlation between the two variables was zero.

I.e. the two variables are independent – the level of rain has no impact on output.

But we can still construct a regression model. And use it to make predictions.



R determines that the model is $\text{output} = 257.4 + 2.316 \text{ rain}$

This suggests that every extra 1 mm of rain contributes on average, 2.316 extra units to output.

But the correlation analysis suggests that the two variables are independent.

So how do we reconcile this?

Statistically significant regression coefficients

Recall the test of correlation ($r=0$) for rain and output

```
> cor.test(rain,output) # p = 0.627
```

Note that $p = 0.627 > 0.05$, which tells us that the variables are independent

We repeat the regression analysis but now pass the output to `summary( )`

```
> model = lm(output ~ rain)
> model
```

```
Call:
lm(formula = output ~ rain)
```

```
Coefficients:
(Intercept)      rain
  257.390       2.316
```

This is the same as before, we just store the output in a variable called *model*, which we now pass to `summary()`

```
> summary(model)
```

```
Call:
lm(formula = output ~ rain)
```

```
Residuals:
    1     2     3     4     5     6     7
-2.022  25.662  10.294   8.346 -8.971  -1.287 -32.022
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  257.390     16.487   15.611 1.96e-05 ***
rain          2.316       4.475    0.518  0.627
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 19.73 on 5 degrees of freedom
Multiple R-squared:  0.05084,    Adjusted R-squared:  -0.139
F-statistic: 0.2678 on 1 and 5 DF,  p-value: 0.6269
```

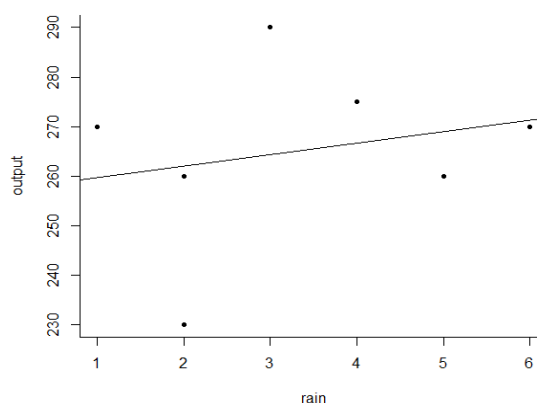
The output includes a table of coefficients – in the last column of this table is the p value for a test that that coefficient is 0.

The p value for the coefficient of *rain* = $0.627 > 0.05$ so this coefficient is not different from zero. I.e. the change in *output* per 1 mm change in rainfall is zero – the two variables are independent!

Another way to interpret the coefficient of *rain* being zero is to remember that it can be interpreted as a slope.

```
> model = lm(output ~ rain)
> plot(rain, output)
> abline(model)
```

A slope of zero is a horizontal line.
And a horizontal line indicates that *output* does not change as *rain* does.



Note that the p value for independent variable in the simple regression model (a scale response regressed on a scale predictor) will be the same as the p value for the correlation test – in this case 0.627

Note that the intercept *is* statistically significant – but this is rarely of interest.

Screen time v exam score again

```
> summary(lm(exam~s.time))
```

Call:

```
lm(formula = exam ~ s.time)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.6167	-2.0369	0.2768	1.0934	4.6200

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	78.2498	2.0945	37.359	2.56e-09 ***
s.time	-7.0532	0.8556	-8.244	7.52e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.3 on 7 degrees of freedom

Multiple R-squared: 0.9066, Adjusted R-squared: 0.8933

F-statistic: 67.96 on 1 and 7 DF, p-value: 7.521e-05

Here we see that the coefficient *s.time* is statistically significant - $p = 7.52 \times 10^{-5}$, the same value we found in the correlation test.

There is some additional useful output generated.

Multiple R-squared: 0.9066 - this is r^2 so we can account for 91% of the variation in exam scores by considering just screen time.

There is also some statistics provided on the residuals, or errors, in the model.

Residuals are the differences between what is observed in the data, and what the model predicts.

Example

The third child in our data spends 2.5 hours a week looking at screens and scored 56 in the exam.

The regression model is $\text{exam} = 78.3 - 7.1 \text{ s.time}$

So, for an $\text{s.time} = 2.5$, the model would predict $78.3 - 7.1 (2.5) = 60.55$

The difference between observed and predicted is $56 - 60.55 = -4.55$. This is the residual for this datum.

Visually, the residual is the vertical distance from the observed data to the regression line.



R reports a 5-point summary for the residuals

I.e. All of the residuals (absolute values) are less than 4.62 etc.

Example 4

For the [lifestyle data](#) (*exercise – asks if respondents get regular exercise or not, the other variables are self-explanatory*), predict the weight of an individual of height 165 cm and discuss the credibility of the prediction.

We begin with a scatter plot to see if a straight-line relationship seems to be a reasonably model.

```
> # alternative to plot(height, weight)
> plot(weight~height)
> abline(lm(weight~height))
```

A linear relationship, albeit, a noisy one is evident.

```
> summary(lm(weight~height))
```

```
Call:
lm(formula = weight ~ height)
```

Residuals:

Min	1Q	Median	3Q	Max
-18.762	-6.195	-1.019	6.145	33.415

Coefficients:

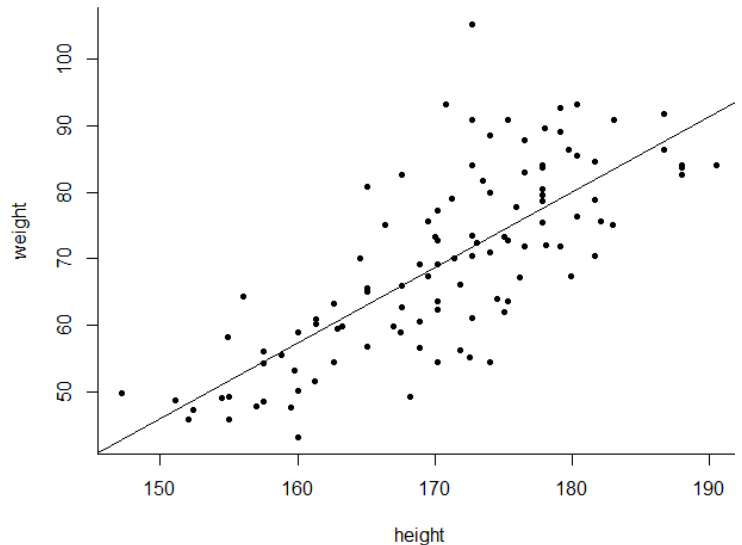
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-124.46662	16.34004	-7.617	1.14e-11 ***
height	1.13637	0.09561	11.886	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.114 on 106 degrees of freedom

Multiple R-squared: 0.5713, Adjusted R-squared: 0.5673

F-statistic: 141.3 on 1 and 106 DF, p-value: < 2.2e-16



The regression equation is $\text{weight} = -124.5 + 1.13 \text{ height}$

Thus, people of height 164 cm will, on average weigh $-124.5 + 1.13 (165) = 61.95$ kg

Thus, our prediction is 61.95 kg.

Discussion

The p value for the height coefficient 1.13, is < 0.05 so height is a meaningful predictor of weight.

I.e. weight is not independent of height.

So, every extra cm in height will see a person's weight increase by an average of 1.13Kg.

How good an estimate is it?

$r^2 = 57\%$ height explains 57% of the variation in weights

50% of the absolute values of the residuals are less than 6.2 Kg.

The value of height for which we are predicting weight lies in the middle of the observed data.

Thus, we are interpolating to make our prediction, which is more credible than extrapolation.

Example

For the customer survey data, (a) quantify the impact (if any) age has on the amount an individual spends and (b) decide if there is any evidence to suggest that the amount a customer spends the quicker their goods will be delivered.

- (a) We begin with a scatter plot to see if a straight-line relationship seems to be a reasonably model.

```
> plot(q4_spend~q2_age)
> abline(lm(q4_spend~q2_age))
```

A linear relationship, albeit, a noisy one is evident.

```
> summary(lm(q4_spend~q2_age))
```

(The output is truncated here)

Call:
lm(formula = q4_spend ~ q2_age)

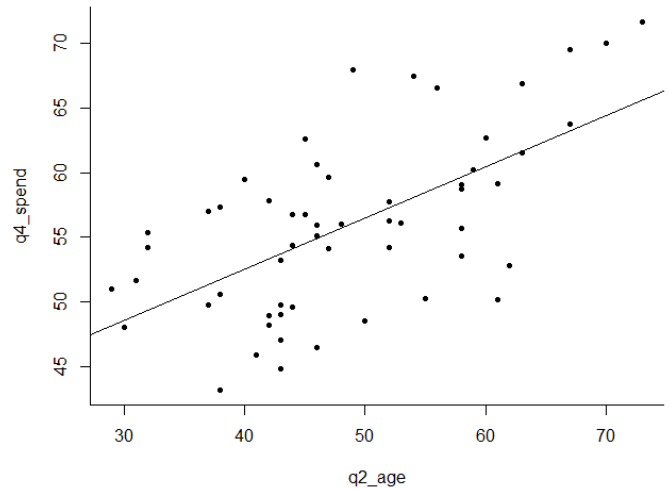
Residuals:

Min	1Q	Median	3Q	Max
-10.6541	-4.0743	0.0645	4.6841	11.9256

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	36.57803	3.43399	10.652	6.98e-15 ***
q2_age	0.39748	0.06904	5.757	4.17e-07 ***

Multiple R-squared: 0.3804, Adjusted R-squared: 0.3689



The output tells us that there is a statistically significant association (i.e. correlation) between *age* and *spend*. The regression equation is $spend' = 36.578 + 0.39748 \text{ age} \sim spend' = 36.6 + 0.4 \text{ age}$

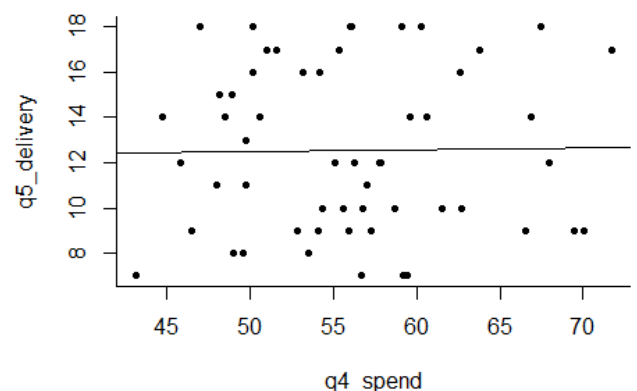
The coefficient of age is 0.4, so the average spend increases by €0.40 or 40c for every extra year of age. So, for example, the average spend for 50-year old customers will be about $36.6 + 0.4 (50) = €56.60$ and this is €4 more than for 40-year olds, whose average spend will be $36.6 + 0.4 (40) = €52.60$

Customer age explains just 38% of variation in spend so there are other factors that might be taken in to account. The predicted spend for half of the customers in the sample is within €5 of their actual spend (Exercise: how can this be determined from the output?)

- (b) A scatter plot of *delivery* v *spend* and a fitted regression line suggest that the variables are independent.

```
> plot(q5_delivery~q4_spend,pch=20, bty="L")
> abline(lm(q5_delivery~q4_spend))
```

It is left as an exercise to show that the p value for the coefficient of spend in the linear model (or, equivalently, the p value for the test of zero correlation between spend and delivery) is greater than 0.05.



Thus, there is no evidence that deliveries are quicker (or slower for that matter) for greater spends.